



Department of Electrical Engineering
EEE3094 Control Systems Engineering
Chapter 3: Laplace Transform revision and Transfer Functions

3. Laplace Transform revision and Transfer Functions

Franklin, Powell and Emami-Naeini (4 th ed.)		Nise (4 th ed.)
Laplace Transforms	Chapter 3.1 & Appendix A	State Space to Transfer Functions: Chapter 3
Dynamic Response:	Chapter 3.4 & 3.5	Time Domain Response: Chapter 4
Bode Plots	Chapter 6.1	Bode Plots: Chapter 10.1 and 10.2

3.1. Laplace Transform revision

This is covered in many textbooks so only a summary will be provided here. We will use the single sided Laplace transform, $Y(s) = \int_{0^-}^{\infty} e^{-s\tau} y(\tau) d\tau = \mathcal{L}\{y(t)\}$. Notice that the negative time component of $y(t)$ does not affect the Laplace transform. Impulses at the origin are captured as the definition used includes the interval $[0^-, 0^+]$. A signal has a Laplace transform with abscissa of convergence $\sigma = \text{Re}\{s\}$ if $\lim_{R \rightarrow \infty} \int_{0^-}^R e^{-st} y(t) dt$, exists $\forall s | \text{Re}\{s\} \geq \sigma$.

It is worth remembering that Laplace transforms are transforms of *signals* and that transfer functions represent the Laplace properties of *derivative operators*.

3.1.1 Laplace transforms of signals: (Use tables, Wolfram alpha or similar sites)

$y(t)$	$Y(s)$	Comments
$\sigma(t)$	$\frac{1}{s}$	unit step
$t\sigma(t)$	$\frac{1}{s^2}$	unit ramp
$e^{-at}\sigma(t)$	$\frac{1}{(s+a)}$	decaying exponential if $a > 0$
$t^n e^{-at}\sigma(t)$	$\frac{n!}{(s+a)^{(n+1)}}$	n^{th} order exponential
$\sin(\omega t)\sigma(t)$	$\frac{\omega}{s^2 + \omega^2}$	sinusoid
$\cos(\omega t)\sigma(t)$	$\frac{s}{s^2 + \omega^2}$	cosinusoid
$e^{-at}\sin(\omega t)\sigma(t)$	$\frac{\omega}{(s+a)^2 + \omega^2}$	damped sinusoid
$e^{-at}\cos(\omega t)\sigma(t)$	$\frac{s+a}{(s+a)^2 + \omega^2}$	damped cosinusoid

3.1.2 Laplace transform properties

Initial value theorem	$\lim_{t \rightarrow 0} y(t) = \lim_{s \rightarrow \infty} sY(s)$
Final value theorem	$\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} sY(s)$
Linearity	$\mathcal{L}\{ax(t) + by(t)\} = a\mathcal{L}\{x(t)\} + b\mathcal{L}\{y(t)\}$
Convolution (with signals zero in negative time)	$\mathcal{L}\left\{\int_0^t x(t-\tau)y(\tau)d\tau\right\} = \mathcal{L}\left\{\int_0^\infty x(t-\tau)y(\tau)d\tau\right\} = X(s)Y(s)$
Translation	$\mathcal{L}\{y(t-T)\} = e^{-sT}Y(s)$
Transform of derivatives	$\mathcal{L}\left\{\frac{d}{dt}y(t)\right\} = sY(s) - y(0)$ $\mathcal{L}\left\{\frac{d^2}{dt^2}y(t)\right\} = s^2Y(s) - sy(0) - y^{(1)}(0)$ $\mathcal{L}\left\{\frac{d^n}{dt^n}y(t)\right\} = s^nY(s) - s^{n-1}y(0) - s^{n-2}y'(0) \dots - y^{(n-1)}(0)$
Transform of integrals	$\mathcal{L}\left\{\int_0^t y(\tau)d\tau\right\} = \frac{1}{s}\left(Y(s) + \int_{-\infty}^0 y(\tau)d\tau\right)$

3.1.3 Inverse Laplace Transform

Formally the inverse transform is found by an integral over a closed contour, C , in the complex s -plane,

$$\mathcal{L}^{-1}\{Y(s)\} = \frac{1}{2\pi j} \oint_C Y(s)e^{st} ds$$

More realistically, for simple functions, we write partial fraction expansions and then use tables.

3.2 Input-Output Descriptions - Transfer Function of Linear Systems

Very often, we are interested in the external behaviour of a system and the state variables are not important to the problem on hand. Obviously the external behaviour cannot depend on the choice of state variables (for example the outflow from a parallel sided tank under gravity with a square-root flow law does not depend on whether we use head, volume or mass as the state variable). This means that the state space description for a given system is not unique. There are *observer* and *controller* canonical forms that are unique but both canonical forms have the same external behaviour and the definition is somewhat arbitrary (you could for instance renumber the state variables). We define the “transfer function”, $P(s)$ of a system as the relationship between the Laplace transform of the input signal and the Laplace transform of the output signal assuming zero initial conditions.

$$Y(s) = P(s)U(s)|_{\text{zero i.c.s}} \quad (3.1)$$

For single input, single output (SISO) systems

$$P(s) = Y(s)/U(s)|_{\text{zero i.c.s}} \quad (3.2)$$

For any state space description, (the S is a compact form of the state space matrices, not to be confused with s , the Laplace variable and later, S , the “sensitivity”!)

$$S = \begin{pmatrix} A & \vdots & B \\ \dots & \cdot & \dots \\ C & \vdots & D \end{pmatrix}$$

$$P(s) = C(sI - A)^{-1}B + D \quad (3.3)$$

It is important to note that we can apply the definition of the transfer function literally by actually taking an input signal (with sufficient energy), calculating the corresponding output signal and then dividing the Laplace transforms. Alternatively, (and more usually) we apply the Laplace transform to the describing differential equation.

We could get to the transfer function by eliminating the state variables state space model model, we end up with an n^{th} order differential equation model,

$$y^{(n)} + a_{n-1}y^{(n-1)} + \dots a_2\ddot{y} + a_1\dot{y} + a_0y = b_mu^{(m)} + b_{m-1}y^{(m-1)} + \dots + b_1\dot{u} + b_0u$$

Taking the Laplace transform with zero initial conditions of the signals and all the derivatives

$$(s^n + a_{n-1}s^{n-1} + \dots a_2s^2 + a_1s + a_0)Y(s) = (b_ms^m + b_{m-1}s^{m-1} + \dots + b_1s + b_0)U(s)$$

Hence, by calculating $P(s) = \frac{Y(s)}{U(s)}$ we get the transfer function.

Motivating Example: Consider the system,

$$\dot{y} + 3y = 2u, \quad y(0) = 0$$

which has transfer function,

$$P(s) = \frac{2}{s+3}$$

If we find the response to a bounded ramp, $u(t) = t\sigma(t) - (t-1)\sigma(t-1)$,

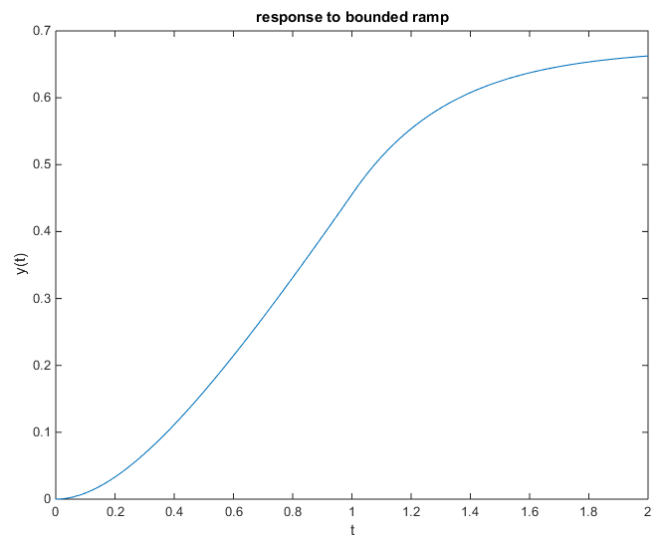
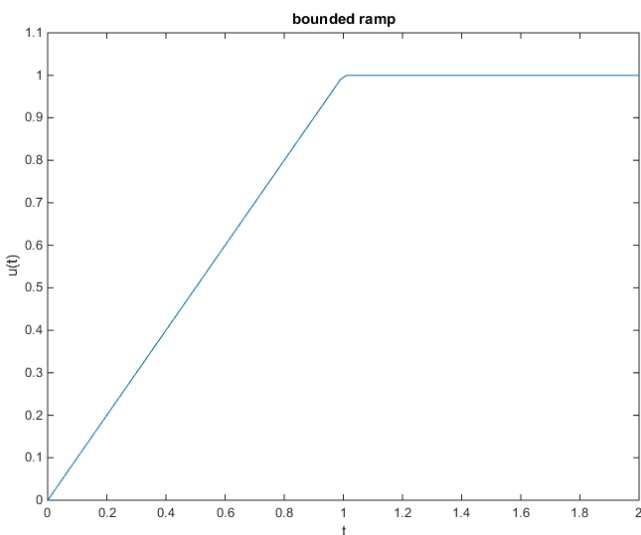
$$U(s) = \frac{1 - e^{-s}}{s^2}$$

This has output,

$$\begin{aligned} Y(s) &= P(s)U(s) = \frac{2(1 - e^{-s})}{(s+3)s^2} \\ &= 2(1 - e^{-s}) \left(\frac{1}{3s^2} - \frac{1}{9s} + \frac{1}{9(s+3)} \right) \end{aligned}$$

and

$$y(t) = 2/9 \left((3t - 1 + e^{-3t})\sigma(t) - (3(t-1) - 1 + e^{-3(t-1)})\sigma(t-1) \right)$$



```
t=linspace(0,2); % produces 100 equally spaced points
u=t-(t-1).*(t>=1);
y=2/9*(3*t-1+exp(-t*3));
y(51:100)=y(51:100)-y(1:50); % for t>=1
plot(t,u)
title('bounded ramp'), ylabel('u(t)'), xlabel('t')
figure, plot(t,y)
title('response to bounded ramp'), ylabel('y(t)'), xlabel('t')
```

Motivating Example Continued: Consider the same system $P(s) = \frac{2}{s+3}$. Take an input signal $u(t) = A \sin \omega t$, which has $U(s) = \frac{A\omega}{s^2 + \omega^2}$ and zero initial conditions (for definition of the transfer function to be valid).

The output is

$$\begin{aligned}
 Y(s) &= P(s)U(s) \\
 &= \frac{2}{s+3} \frac{A\omega}{s^2 + \omega^2} \\
 &= \frac{2A\omega}{9 + \omega^2} \left(\frac{1}{s+3} + \frac{-s+3}{s^2 + \omega^2} \right) \\
 y(t) &= \frac{2A\omega}{9 + \omega^2} \left(e^{-3t} + \left(-\cos \omega t + \frac{3}{\omega} \sin \omega t \right) \right) \\
 &= \frac{2A\omega}{9 + \omega^2} e^{-3t} + \frac{2A}{\sqrt{9 + \omega^2}} \sin \omega t + \phi, \quad \phi = \text{atan}(-\omega/3)
 \end{aligned}$$

(recall $X \sin \theta + Y \sin \theta = \sqrt{X^2 + Y^2} \sin(\theta + \text{atan } Y/X)$. In code, use the two argument atan: `atan2(Y,X)`.)

The first term is the transient response, and the second (which is of interest at present) is the harmonic response.

The harmonic response is a sinusoid which has an amplitude, $A|P(j\omega)|$, and a phase, $\arg P(j\omega)$, ($j = \sqrt{-1}$).

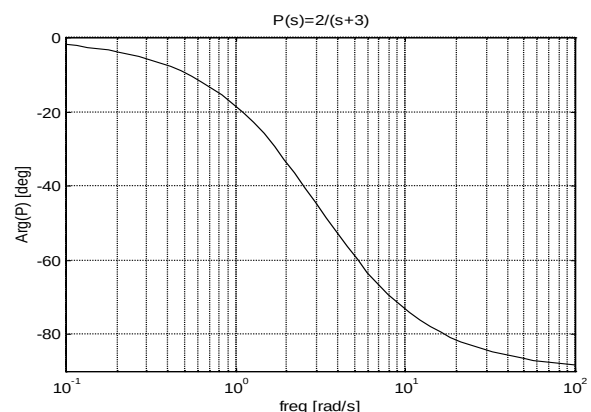
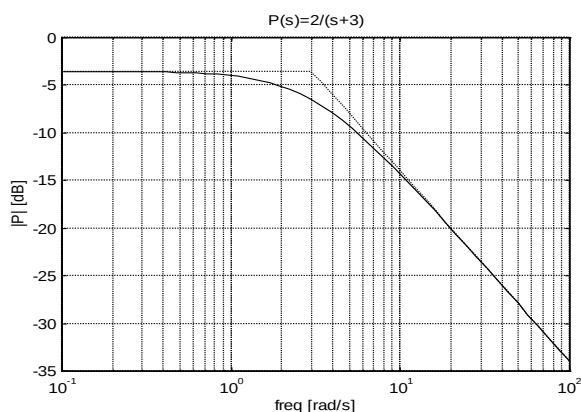
$$P(j\omega) = \frac{2}{j\omega + 3} = \frac{2}{\sqrt{\omega^2 + 3^2}} e^{-j \text{atan} \omega/3}$$

We could have obtained the same result using phasor methods (from an Electricity course) or using the Fourier transform (which is an integral defined over all time and therefore would not have had the problem with the initial conditions).

The transfer function calculated with $s = j\omega$ is a very useful tool in understanding system behaviour.

Now draw the Bode plot of $P(s) = \frac{2}{s+3}$ above, with $P(j\omega) = \frac{2}{j\omega + 3} = \frac{2}{\sqrt{\omega^2 + 3^2}} e^{-j \text{atan} (\omega/3)}$.

ω	$ P(j\omega) $	$ P(j\omega) _{\text{dB}}$	$\arg P(j\omega)$
$\omega \ll 3$	$\approx 2/3$	$20 \log_{10} 2/3 = -3.52 \text{ dB}$	$-\text{atan } \omega/3 \approx 0^\circ$
$\omega \gg 3$	$\approx 2/\omega$	$20 \log_{10} 2 - 20 \log_{10} \omega$	$-\text{atan } \omega/3 \approx -90^\circ$
$\omega \approx 3$	$\approx 2/(3\sqrt{2})$	$20 \log_{10} 2/3 - 20 \log_{10} \sqrt{2}$ $= -3.52 - 3.01 = -6.53 \text{ dB}$	$-\text{atan } \omega/3 \approx -45^\circ$



Aside for above:

$$\begin{aligned}\sin A + B &= \sin A \cos B + \cos A \sin B \\ X \sin \theta + Y \cos \theta &= \sqrt{X^2 + Y^2} \left(\frac{X}{\sqrt{X^2 + Y^2}} \sin \theta + \frac{Y}{\sqrt{X^2 + Y^2}} \cos \theta \right) \\ &= \sqrt{X^2 + Y^2} \sin(\theta + B) \\ \cos B &= \frac{X}{\sqrt{X^2 + Y^2}}, \quad \sin B = \frac{Y}{\sqrt{X^2 + Y^2}} \Rightarrow \tan B = \frac{\sin B}{\cos B} = \frac{Y}{X}\end{aligned}$$

3.3 Matlab

Note: Matlab has several utilities for doing routine calculations used in this chapter. In addition to being able to do these calculations by hand for small problems, you should be able to use the following.

Command	Function
residue	Partial-fraction expansion or residue computation
ss2tf	State-space to transfer function conversion
tf2ss	Transfer function to state-space conversion
s = tf('s')	Define a symbolic Laplace variable
P=tf((2*s+3)/(s^2+4*s+1))	Define the transfer function P
P=tf([2, 3],[1 4 1])	Equivalent to command above, using the numerator and denominator polynomial coefficient vectors
eig	Eigenvalues and eigenvectors.
roots	Polynomial roots.
step	Step response.
bode	Bode plot (frequency response)
logspace	Logarithmically spaced vector

3.3: Step responses and Bode Plots

(Nise, Chapter 4; Franklin, Powell and Emani-Naeini Sections 3.3-3.5)

Although actual plant signals are often not steps, the unit step response is often used as a generic performance specification for systems. $u(t) = \sigma(t)$, $U(s) = 1/s$. Notice that $1/s$ has energy at all frequencies whereas a sinusoid only excites a single frequency.

3.3.1 First order low-pass systems

$P(s) = \frac{k}{s/a+1}$, then has a step response, $y(t) = k(1 - e^{-at})$, where

k is the steady state gain;

$a = 1/\tau$ is the corner frequency; and

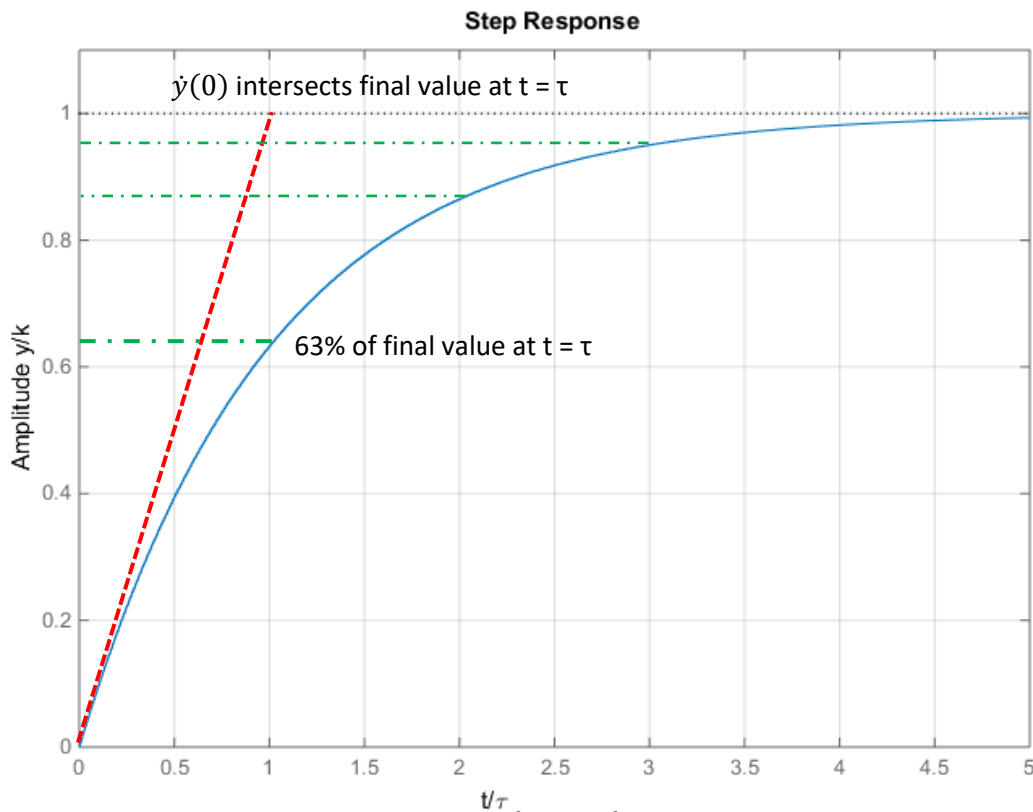
τ is the time constant

$$P(s) = \frac{k}{s/a+1}, u(t) = \sigma(t), U(s) = \frac{1}{s}, Y(s) = P(s)U(s) = \frac{k}{(s/a+1)s} = \frac{k}{s} - \frac{k}{s+a}, y(t) = k(1 - e^{-at})$$

Step response of first order low pass term

The step response is annotated below. Points of support for the sketch include the fact that

- $y(t) = k(1 - e^{-1}) = 0.63k$, at $t = \tau = 1/a$, and the slope of the response is,
- $\dot{y}(t) = kae^{-at} = ka$ at $t = 0$
- The response gets to 14% and 5% of the final value after 2 and 3 time constants respectively.



Bode plot of first order low pass term

$$P(j\omega) = \frac{k}{j\omega/a + 1}$$

$$|P(j\omega)|_{dB} = 20 \log_{10} k - 20 \log_{10} \sqrt{(\omega/a)^2 + 1}$$

$$\angle P(j\omega) = -\text{atan2}(\omega/a, 1)$$

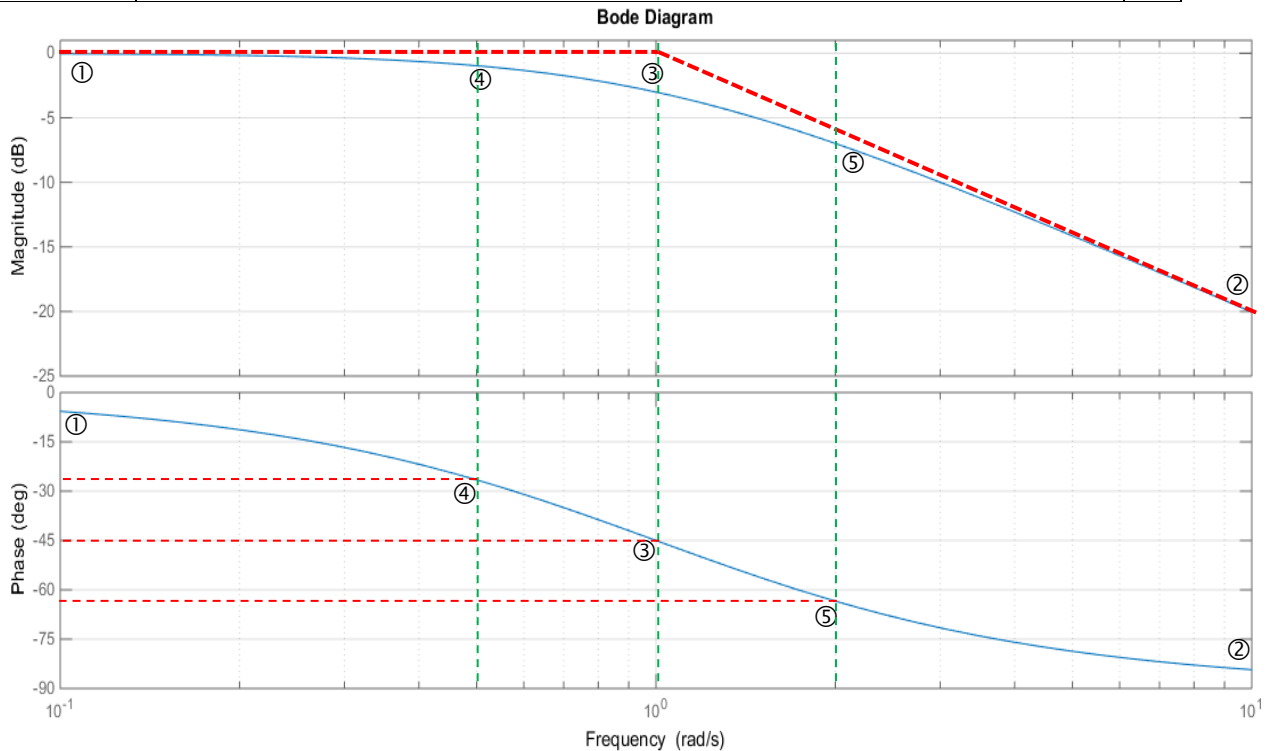
(Note: atan2(y,x) gives the angle of x+iy in the correct quadrant.)

Magnitude plot: The following asymptotic values apply

$\omega \ll a$	$ P(j\omega) _{dB} \approx k_{dB}$	①
$\omega \gg a$	$ P(j\omega) \approx k/\omega$, $ P(j\omega) _{dB} \approx k_{dB} - 20 \log_{10}(\omega/a)$ $ P $ rolls off at -20 dB/decade or -6 dB/octave (i.e. doubling of frequency)	②
$\omega = a$	$ P(j\omega) = k/\sqrt{2}$ $ P(j\omega) _{dB} = k_{dB} - 3 \text{ dB}$	③
$\omega = a/2$	$ P(j\omega) = k/\sqrt{1.25}$ $ P(j\omega) _{dB} = k_{dB} - 1 \text{ dB}$ (at half the corner frequency)	④
$\omega = 2a$	$ P(j\omega) = k/\sqrt{5}$ $ P(j\omega) _{dB} = k_{dB} - 6 - 1 \text{ dB}$ (at double the corner frequency)	⑤

Phase plot: The following asymptotic values apply

$\omega \ll a$	$\arg(P(j\omega)) \approx 0^\circ$	①
$\omega \gg a$	$\arg(P(j\omega)) \approx -90^\circ$	②
$\omega = a$	$\arg(P(j\omega)) = -\text{atan}(1) = -45^\circ$	③
$\omega = a/2$	$\arg(P(j\omega)) = -\text{atan}(1/2) \approx -27^\circ$	④
$\omega = 2a$	$\arg(P(j\omega)) = -\text{atan}(2) \approx -63^\circ$ ($= -90^\circ + 27^\circ$)	⑤



Bode plot with $k = 1$, $a = 1$ (alternatively, frequency axis is ω/a)

3.3.2 Second Order Low Pass System

$$T_{low} = \frac{k}{(s/\omega_n)^2 + 2\zeta s/\omega_n + 1}$$

Note that the numerator order is 0. The high frequency roll-off is 2nd order.

Step response of 2nd order low pass system

Check that you can get the response:

$$y(t) = k \left(1 - e^{-\sigma t} \left(\frac{\zeta}{\sqrt{1-\zeta^2}} \sin(\omega_d t) + \cos(\omega_d t) \right) \right) = k \left(1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\sigma t} \sin(\omega_d t + \phi) \right)$$

Notation:

ζ is the *damping factor*. For $\zeta \geq 1$, the system is *over-damped* and the denominator polynomial can be factorised into two first order terms (with real coefficients).

ω_n is the *undamped natural frequency*. If $\zeta = 0$, the system is *undamped* and will oscillate at ω_n rad/s forever (i.e. not damped).

$\omega_d = \omega_n \sqrt{1-\zeta^2}$ is the damped natural frequency. (Only makes sense if $0 < \zeta < 1$.)

$$\phi = \arccos(\zeta) = \arctan\left(\frac{\zeta}{\sqrt{1-\zeta^2}}\right)$$

$\sigma = \zeta \omega_n$. $1/\sigma$ is often called the *time constant* of the second order system.

Solution of step response for second order low passing transfer function,

$$T_{low} = \frac{k}{(s/\omega_n)^2 + 2\zeta s/\omega_n + 1}$$

The input signal is,

$$u(t) = \sigma(t) - \text{unit step}, \quad U(s) = \frac{1}{s}$$

The output signal Laplace transform is the product of the transfer function and the Laplace transform of the input,

$$\begin{aligned} Y(s) &= T_{low}(s) U(s) \\ &= \frac{k\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)} \\ &= \frac{A}{s} + \frac{Bs + C}{s^2 + 2\zeta\omega_n s + \omega_n^2} \end{aligned}$$

Partial fraction expansion – numerator terms:

$$A(s^2 + 2\zeta\omega_n s + \omega_n^2) + Bs^2 + Cs = k\omega_n^2$$

$$\left. \begin{aligned} s^0: & \quad A\omega_n^2 = k\omega_n^2 \\ s^1: & \quad A2\zeta\omega_n + C = 0 \\ s^2: & \quad A + B = 0 \end{aligned} \right\} \begin{aligned} A &= k \\ C &= -k2\zeta\omega_n \\ B &= -k \end{aligned}$$

Output is therefore,

$$\begin{aligned} Y(s) &= k \left(\frac{1}{s} - \frac{s + 2\zeta\omega_n}{s^2 + 2\zeta\omega_n s + \omega_n^2} \right) \\ &= k \left(\frac{1}{s} - \frac{(s + \zeta\omega_n) + \zeta\omega_n}{(s + \zeta\omega_n)^2 + (1-\zeta^2)\omega_n^2} \right) \end{aligned}$$

The relevant Laplace transforms are below

Time Domain Signal	Laplace Transform
$\sigma(t)$	$\frac{1}{s}$
$e^{-at} \sin \omega t \sigma(t)$	$\frac{\omega}{(s+a)^2 + \omega^2}$
$e^{-at} \cos \omega t \sigma(t)$	$\frac{s+a}{(s+a)^2 + \omega^2}$

Defining: $\omega_d = \omega_n \sqrt{1 - \zeta^2}$, the damped natural frequency, $\sigma = \zeta \omega_n$ and noting $\frac{\zeta \omega_n}{\omega_d} = \frac{\zeta}{\sqrt{1 - \zeta^2}}$

$$Y(s) = k \left(\frac{1}{s} - \frac{\zeta \omega_n}{(s + \sigma)^2 + \omega_d^2} - \frac{(s + \sigma)}{(s + \sigma)^2 + \omega_d^2} \right)$$

With inverse Laplace transform,

$$y(t) = k \left(1 - e^{-\sigma t} \left(\frac{\zeta}{\sqrt{1 - \zeta^2}} \sin(\omega_d t) + \cos(\omega_d t) \right) \right)$$

To classify the response, the following support points are used

Time to peak

Time to peak is the turning point of $y(t)$: $t_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}}$, Peak value: $y(t_p) = k \left(1 + e^{-\pi \zeta / \sqrt{1 - \zeta^2}} \right)$

This gives the overshoot, $M_p = \frac{y(t_p) - y(\infty)}{y(\infty)} = e^{-\pi \zeta / \sqrt{1 - \zeta^2}}$.

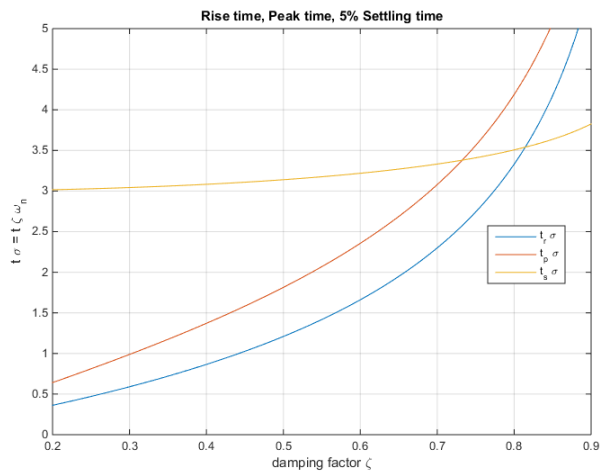
Rise time

The first crossing of the final value is at $\sin(\omega_d t_r + \phi) = 0$, or $t_r = (\pi - \arccos(\zeta)) / (\omega_n \sqrt{1 - \zeta^2})$ (obviously, always less than t_p .) (Nise and others use the rise time to 90% of the final value and then get an expression that does not have a neat (i.e. not nice!) solution.)

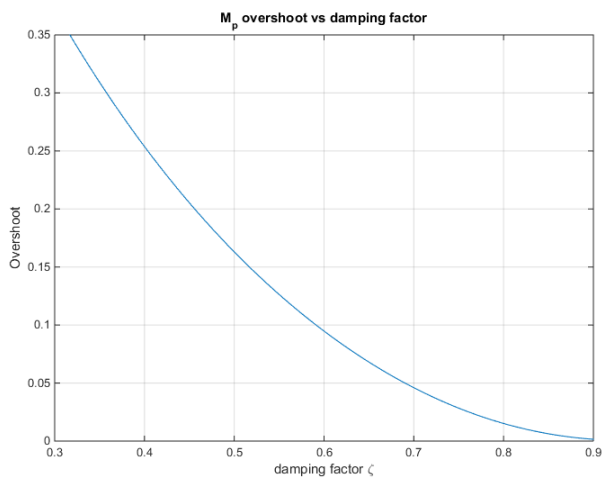
Settling time

There are various definitions depending on whether you use the response, $y(t)$, itself or the envelope, $y_{\text{env}}(t) = k \left(1 \pm \frac{1}{\sqrt{1 - \zeta^2}} e^{-\sigma t} \right)$ (noting that the extreme values of $\sin(\omega_d t + \phi)$ are ± 1). The envelope gets

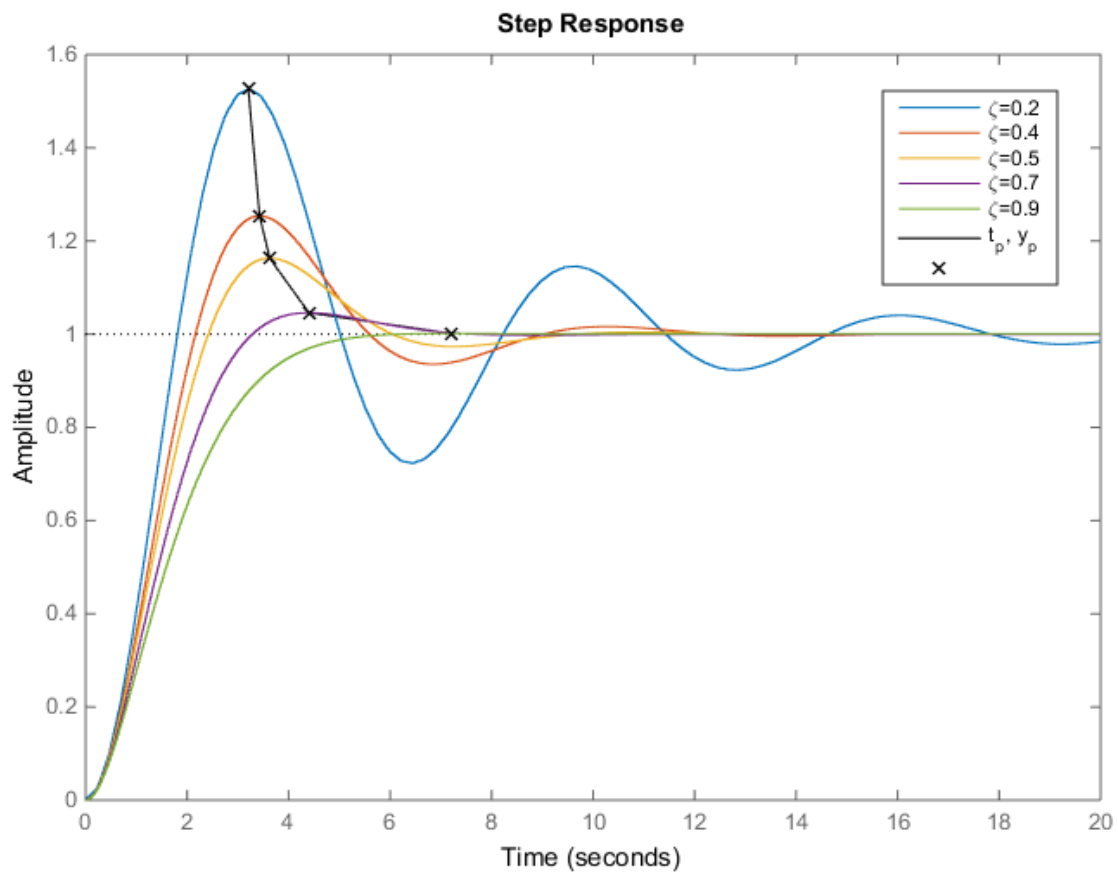
to 5% and 1% as a function of ζ as shown below $t_{s \, x\%} = \frac{-\ln(x\sqrt{1 - \zeta^2})}{\zeta \omega_n}$. (The equivalent answer for the actual response is discontinuous in ζ .)

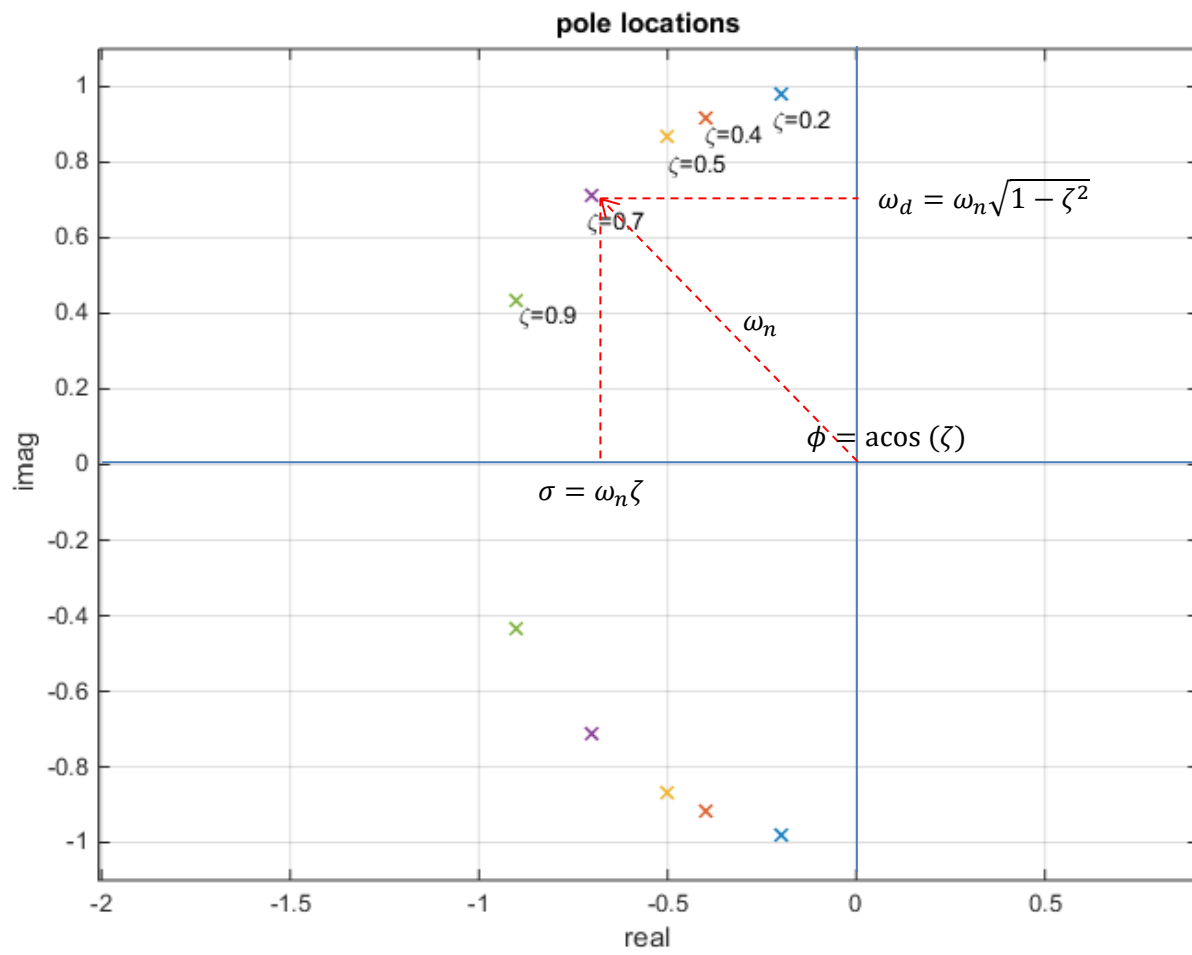


```
% calculate second order stuff
% define damping factor range
zeta=linspace(0.3,0.9);
% rise time * omega_n
tr=(pi-acos(zeta))./sqrt(1-zeta.^2).*zeta;
% peak time * omega_n
tp=(pi)./sqrt(1-zeta.^2) .*zeta;
% 5% settling time
ts5=-log(0.05*sqrt(1-zeta.^2));
plot(zeta,tr,zeta,tp,zeta,ts5),grid
title('Rise time, Peak time, 5% Settling
time')
ylabel('t \omega_n')
xlabel('damping factor \zeta')
shg
```



```
% M_p formula
% define damping factor range
zeta=linspace(0.3,0.9);
plot(zeta,exp(-pi*zeta./sqrt(1-zeta.^2)))
title('Overshoot vs damping factor')
ylabel('Overshoot')
xlabel('damping factor \zeta')
```





Bode plot of 2nd order low-pass system

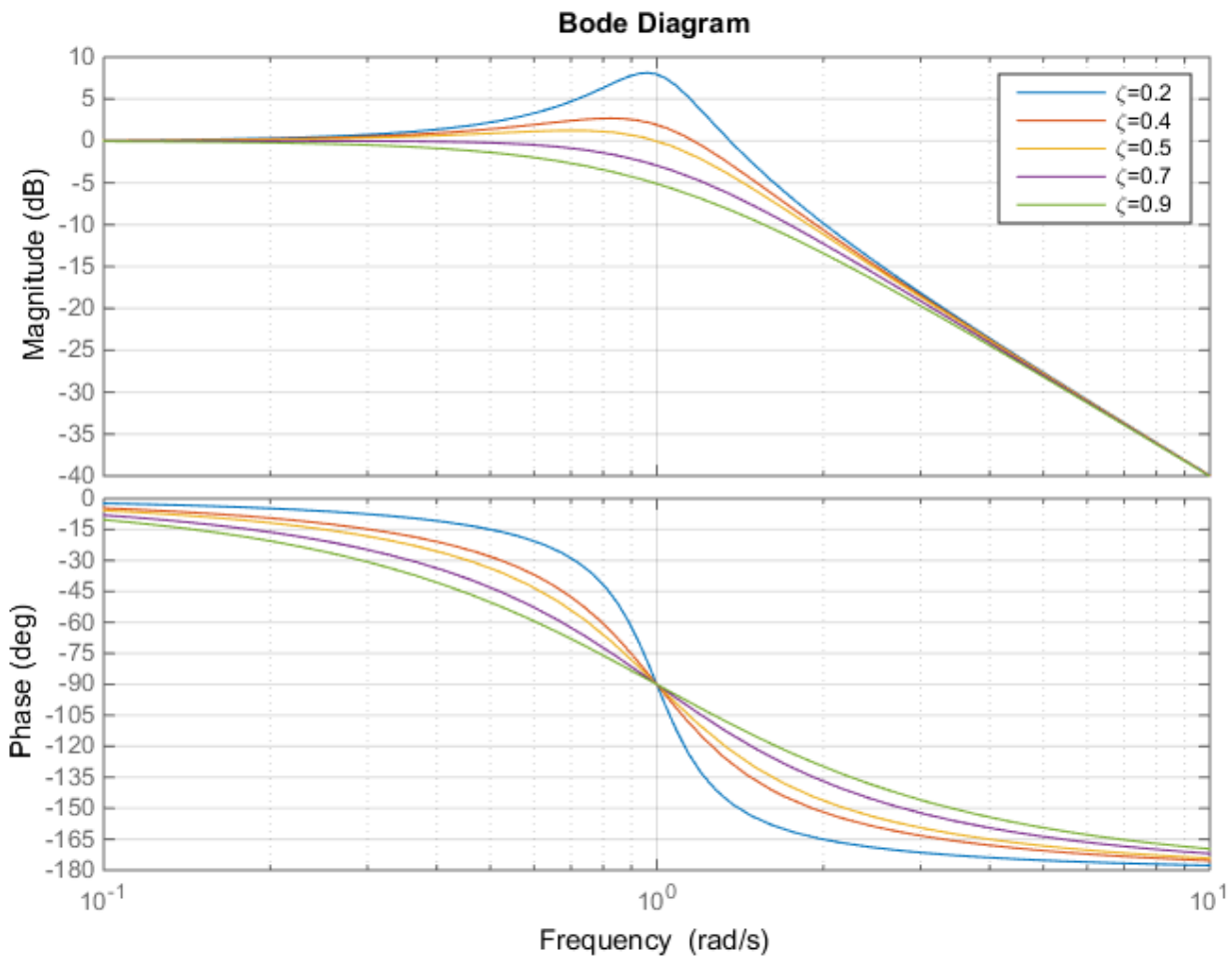
$$T = \frac{k}{(s/\omega_n)^2 + 2\zeta s/\omega_n + 1}, \quad T(j\omega) = \frac{k}{1 - (\omega/\omega_n)^2 + j2\zeta \omega/\omega_n}$$

$$|T(j\omega)| = \frac{k}{\sqrt{(1 - (\omega/\omega_n)^2)^2 + (2\zeta \omega/\omega_n)^2}}$$

$$|T(j\omega)|_{dB} = k_{dB} - 20 \log_{10} \sqrt{(1 - (\omega/\omega_n)^2)^2 + (2\zeta \omega/\omega_n)^2}$$

$$\arg T(j\omega) = \arg k - \tan^{-1} \frac{2\zeta \omega/\omega_n}{1 - (\omega/\omega_n)^2}$$

As above, we get a horizontal low frequency asymptote of k_{dB} and a high frequency asymptote of -40 dB/decade (-12 dB/octave) rolling off from the corner frequency. (The roll off is twice that of a first order term.) The support of the Bode magnitude plot includes the peak, $\omega_m = \omega_n \sqrt{1 - 2\zeta^2}$ with $M(\omega_m) = k_{dB} - 20 \log_{10} 2\zeta \sqrt{1 - \zeta^2}$; and the value at the corner frequency, $M(\omega_n) = k_{dB} - 20 \log_{10} 2\zeta$. Plotted below for $k = 1$ and normalised in frequency (ω/ω_n).



See Appendix for points of support for the phase plot.

3.3.3 Formulas for t_p and y_p for band-pass systems with a unit step input:

$$T_{Y/D} = k \frac{2\zeta s / \omega_n}{(s / \omega_n)^2 + 2\zeta s / \omega_n + 1}$$

For $d(t) = \sigma(t)$, a unit step, $D(s) = 1/s$

$$y(t) = \frac{k2\zeta}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin \omega_d t$$

$\omega_d = \omega_n \sqrt{1-\zeta^2}$ the damped natural frequency.

(NB $\omega_d = \omega_n \sqrt{1-\zeta^2}$ is not the same as the peak frequency $\omega_m = \omega_n \sqrt{1-2\zeta^2}$ above.)

For turning points, $t = t_p$,

$$\frac{dy}{dt} = \frac{k2\zeta}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} (-\zeta\omega_n \sin \omega_d t + \omega_d \cos \omega_d t) = 0|_{t=t_p}$$

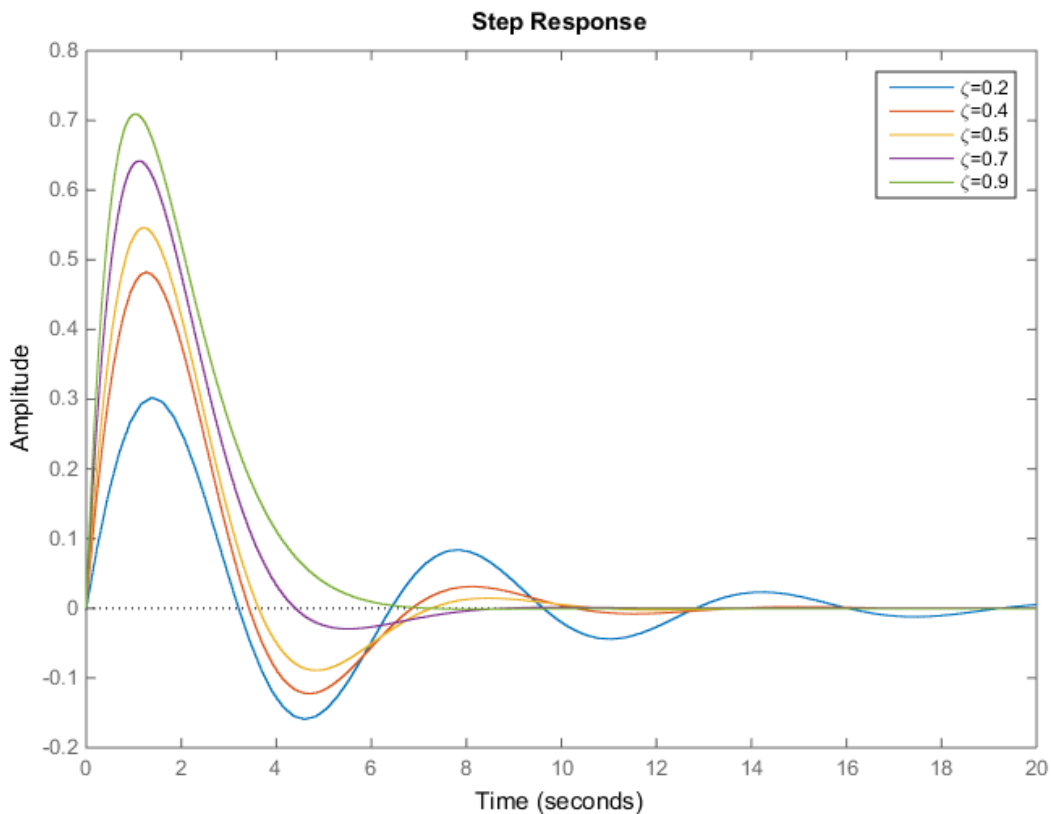
$$\tan \omega_d t_p = \frac{\sqrt{1-\zeta^2}}{\zeta} \quad \text{or} \quad \cos \omega_d t_p = \zeta$$

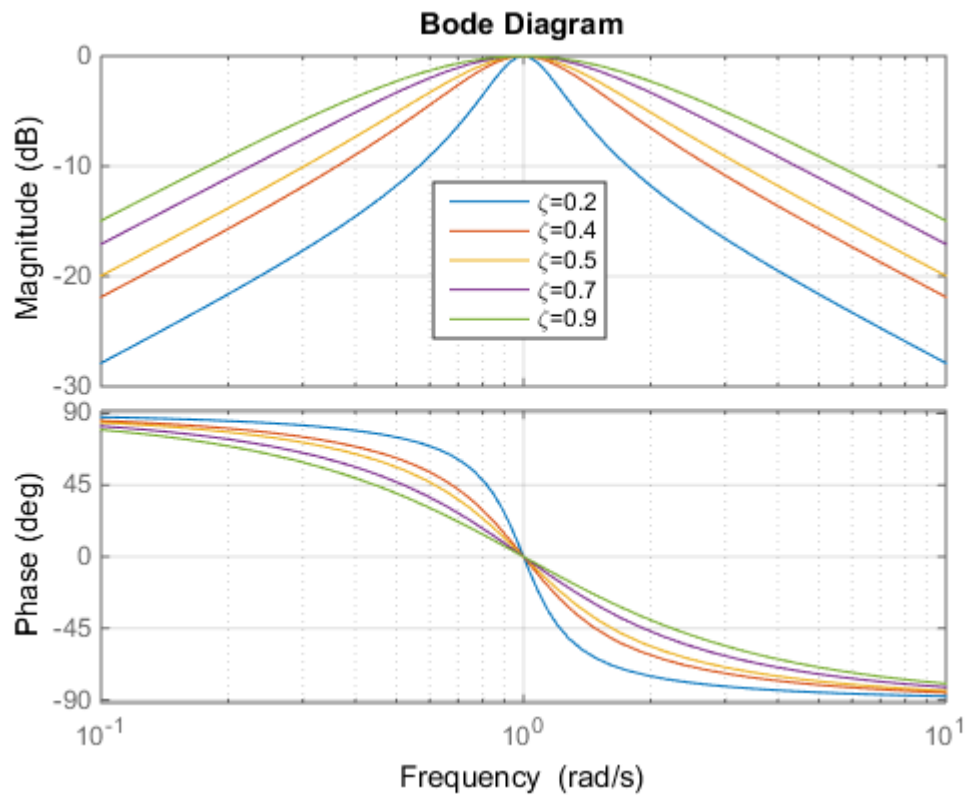
The peak value is

$$y(t_p) = k2\zeta e^{-\zeta \cos \zeta / \sqrt{1-\zeta^2}}$$

which depends only on ζ and k .

$2\zeta e^{-\zeta \cos \zeta / \sqrt{1-\zeta^2}}$ is the number we look up on the normalised charts.



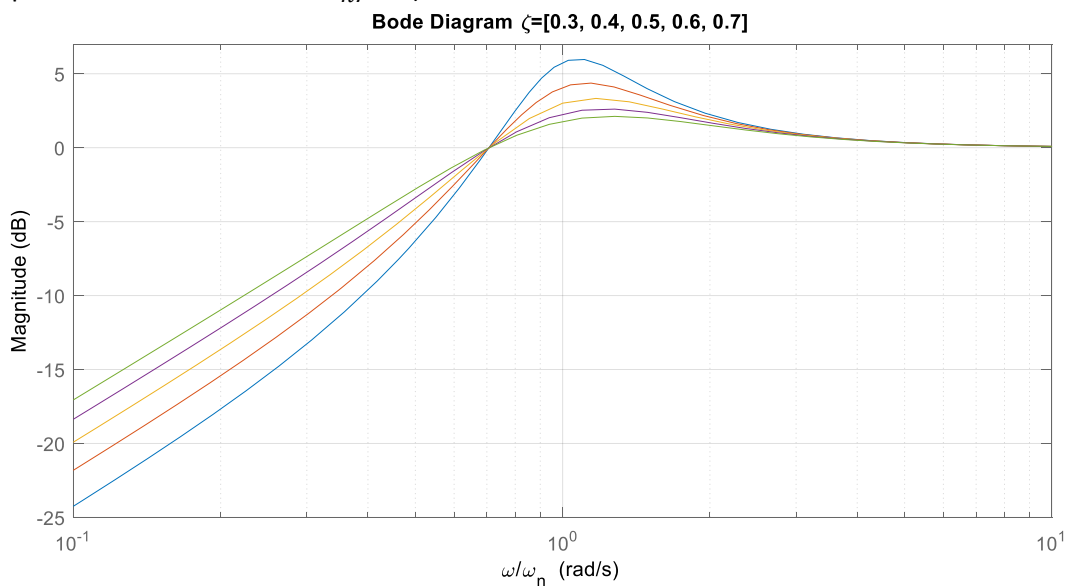


3.3.4 High Pass Formulas

Develop the time-domain formulas from the low pass ones if

$$\begin{aligned}
 T_{high} &= 1 - T_{low} \\
 &= 1 - \frac{1}{(s/\omega_n)^2 + 2\zeta s/\omega_n + 1} \\
 &= \frac{(s/\omega_n)^2 + 2\zeta s/\omega_n}{(s/\omega_n)^2 + 2\zeta s/\omega_n + 1}
 \end{aligned}$$

In the frequency domain, the peak occurs *above* the corner frequency at $\omega_{max} = \omega_n \sqrt{\frac{1+\sqrt{1+2\zeta^2}}{2}}$. (All responses cross 0 dB at $\omega = \omega_n/\sqrt{2}$.)



Develop your own formula sheet with *all* the relevant equations and learn how to apply them!

3.3.5 General time domain response via the Laplace transform

In general, if we have a transfer function,

$$P(s) = \frac{b_{np}s^{np} + \dots + b_1s + b_0}{s^{dp} + a_{dp-1}s^{dp-1} + \dots + a_1s + a_0} = \frac{\sum_{h=0}^{np} b_h s^h}{\sum_{h=0}^{dp} a_h s^h} = \frac{b_{np} \prod_{h=1}^{np} (s + z_h)}{\prod_{h=1}^{dp} (s + p_h)}$$

where $s = -p_h$ are the “poles” (where $P(s) \rightarrow \infty$) and $s = -z_h$ are the “zeros” (where $P(s) \rightarrow 0$). np and dp are the orders of the numerator and denominator polynomials respectively. If there is a second order term, say in the denominator, there is a complex conjugate pair of poles, $p_r = p_{r+1}^*$ in the factorisation. (Any polynomial can be factored into a product of first and second order terms and second order terms can be factorised as $(s + p)(s + p^*) = s^2 + 2\text{Re}\{p\}s + (\text{Re}\{p\}^2 + \text{Im}\{p\}^2)$ using a “completing the square” approach.)

A system is “proper” if, $np \leq dp$, and “strictly proper” if, $np < dp$. If you assume that you can describe an inductor with input as current and output as voltage, you will have a transfer function, $P(s) = sL$ (i.e. $dp = 0$, $np = 1$) corresponding to behaviour, $V(s) = sL I(s)$. This is not proper because if the input is e.g. a step, the output would be an impulse which is not realistic. The natural input to an inductor is the voltage and the current is the output or consequence, i.e. $I(s) = \frac{1}{sL} V(s)$, which is strictly proper.

A general input signal has a Laplace transform, $U(s)$. If we limit our attention (for now) to signals that can be represented by complex exponentials, we have

$$U(s) = \frac{b_{nu} \prod_{h=1}^{nu} (s + z_h)}{\prod_{h=1}^{du} (s + p_h)}$$

And the output is

$$Y(s) = P(s)U(s) = \frac{b_{np} \prod_{h=1}^{np} (s + z_h)}{\prod_{h=1}^{dp} (s + p_h)} \frac{b_{nu} \prod_{h=1}^{nu} (s + z_h)}{\prod_{h=1}^{du} (s + p_h)} = \frac{b_{ny} \prod_{h=1}^{ny} (s + z_h)}{\prod_{h=1}^{dy} (s + p_h)}$$

$dy = dp + du$ and $ny = np + nu$. $Y(s)$ can be written as a partial fraction expansion. In the case of no repeated poles, the result is,

$$Y(s) = \sum_{h=1}^{dy} \left(\frac{r_h}{s + p_h} \right) + k$$

In the case of strictly proper systems, $k = 0$. If there are complex conjugate poles, they will have complex conjugate residues. Systems with repeated roots require more care – see textbooks. Systems with repeated roots are technically relevant: For example, in a jet engine or steam turbine, all the turbine blades are made to be identical and you may need to study complicated modes where the coupled dynamics are excited.

We can find the resulting response of $y(t) = \mathcal{L}^{-1}\{Y(s)\}$ by inverse Laplace transform, term by term;

$$y(t) = \sum_{h=1}^{dy} (r_h e^{-p_h t}) + k$$

(Obviously, $y(t)$ is real as an oscilloscope does not have an imaginary input probe.) Check the Matlab command `respk(...)` that finds residues or allows you to enter system models in partial fraction format.

3.3.6 Bode plots

In control design we will often use the so-called frequency response. If a stable (to be defined below) linear system is excited with a steady sinusoidal input, the output will also be sinusoidal after initial transients have died out. In the 1940's (i.e. before PC's and smart-phones) all calculations were done by hand. Bode worked at Bell Laboratory in the USA on problems related to amplifier performance (before transistors and op amps!). We will discover that there are very useful relationships between the frequency domain and time domain behaviour and the Bode plot is a simple way of representing behaviour in the frequency domain. A Bode plot displays $|P(j\omega)|$ in dB (defined below) and $\text{Arg}(P(j\omega))$ as a function of frequency using semi logarithmic paper so that large frequency range can be represented. (Incidentally the frequency of notes on a piano is logarithmically spaced and eight ivory/white plastic notes make an “octave” which is a doubling of frequency e.g. from C to C.)

Definition - deciBell (dB)

$$\begin{aligned} x_{dB} &= 20 \log_{10} |x| \\ &= 10 \log_{10} |x|^2 \end{aligned}$$

(We remember the first formula but the second shows that the 20 is not arbitrary: 10 = deci and the power 2 indicates that the dB is a measure of power (i^2R , V^2/R etc).)

As a piece of history, the other unit that was used is the neper:
 $|x|_{\text{neper}} = \ln|x|$, and 1 neper is $20 \log_{10} e = 8.69 \text{ dB}$

Discussion

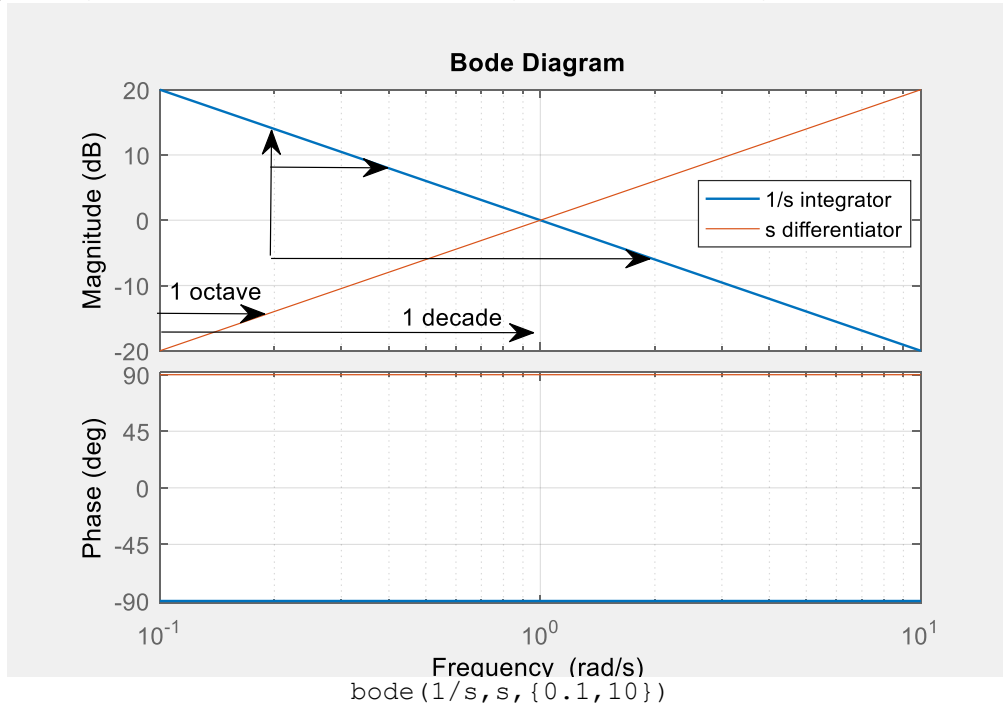
As above, a polynomial can be factored into a product of first and second order terms

$$\begin{aligned} P(s) &= \frac{b_{np}s^{np} + \dots + b_1s + b_0}{s^{dp} + a_{dp-1}s^{dp-1} + \dots + a_1s + a_0} = \frac{\sum_{h=0}^{np} b_h s^h}{\sum_{h=0}^{dp} a_h s^h} \\ &= \frac{b_{np} \prod_{h=1}^{np} (s + z_h)}{\prod_{h=1}^{dp} (s + p_h)} \\ &= b_{np} \frac{\prod_{h=1}^{np1} (s + z_h)}{\prod_{h=1}^{dp1} (s + p_h)} \frac{\prod_{nh=1}^{np2} (s^2 + 2\zeta_{nh}\omega_{nh}s + \omega_{nh}^2)}{\prod_{dh=1}^{dp2} (s^2 + 2\zeta_{dh}\omega_{dh}s + \omega_{dh}^2)} \end{aligned}$$

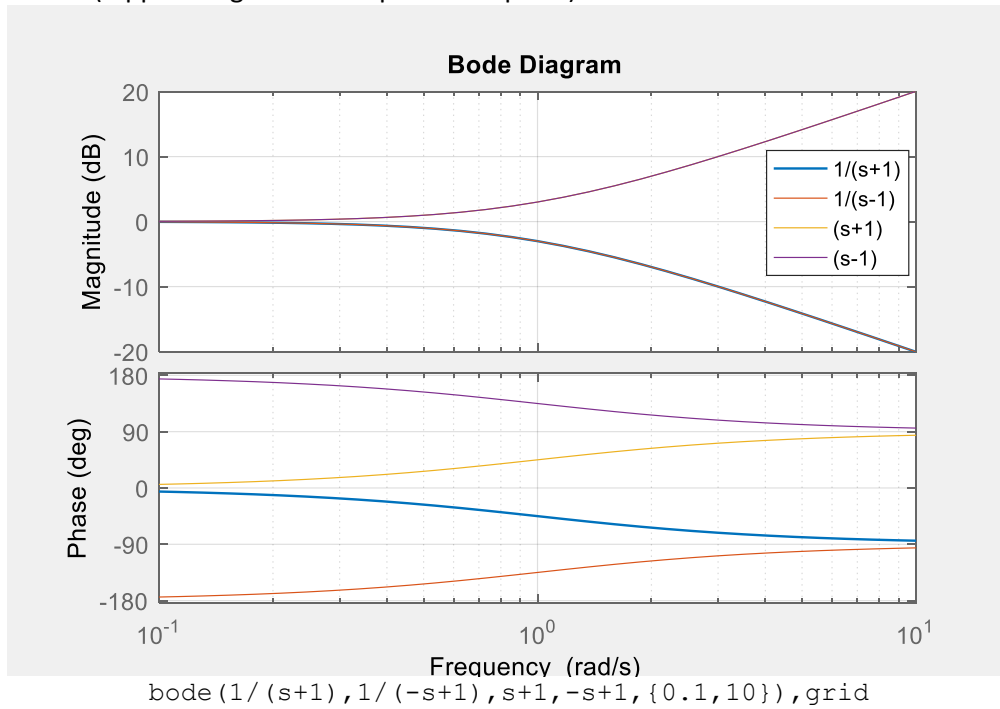
In the last line, the coefficients are all real as we have collected all the complex conjugate pairs. To make sense, $np = np1 + 2 \times np2$ and $dp = dp1 + 2 \times dp2$. This means that we only need to be able to draw first and second order factors and add since multiplication becomes addition in the complex logarithmic plane (dB and degrees).

There are only a few types of terms:

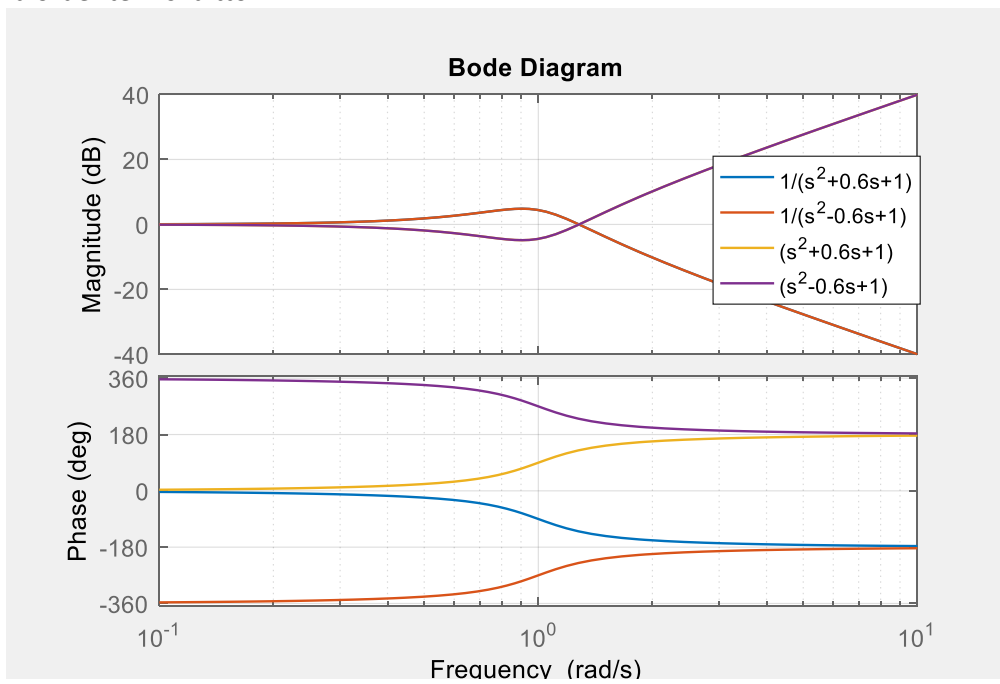
- Integrators (-20 dB/decade = -6 dB/octave, -90°) and differentiators (+20 dB/decade, $+90^\circ$)



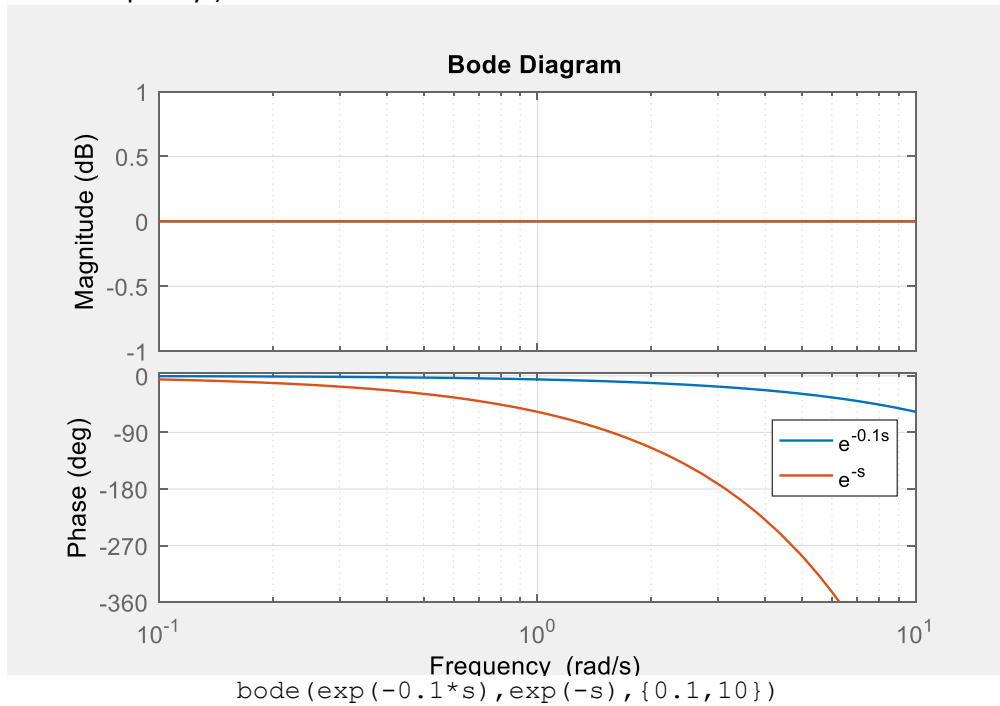
- First order terms: LH plane pole, RH plane pole (same magnitude, flipped phase), LH plane zero and RH plane zero (flipped magnitude and phase wrt poles).



- Second order terms: ditto



- Dead time: Constant magnitude and phase goes exponentially wrt logarithmic frequency (i.e. linearly wrt linear frequency!)



Computers can draw nice Bode plots.

Humans should practice and sketch (± 1 dB, $\pm 5^\circ$) and interpret.

Appendix: Points of support for Bode phase plots.

First order low pass

$$T = 1/(s/a + 1)$$

$$\arg T(j\omega) = -\operatorname{atan}(x), \text{ with } x = \omega/a$$

Slope of the argument with respect to $\ln x$ (as the Bode plot is drawn with respect to $\log \omega$)

$$\begin{aligned} \frac{d \arg T}{d \ln x} &= \frac{d \arg T}{d x} \bigg/ \frac{d \ln x}{d x} \\ &= -\left(\frac{1}{1+x^2}\right) \bigg/ \left(\frac{1}{x}\right) = -\frac{x}{1+x^2} \end{aligned}$$

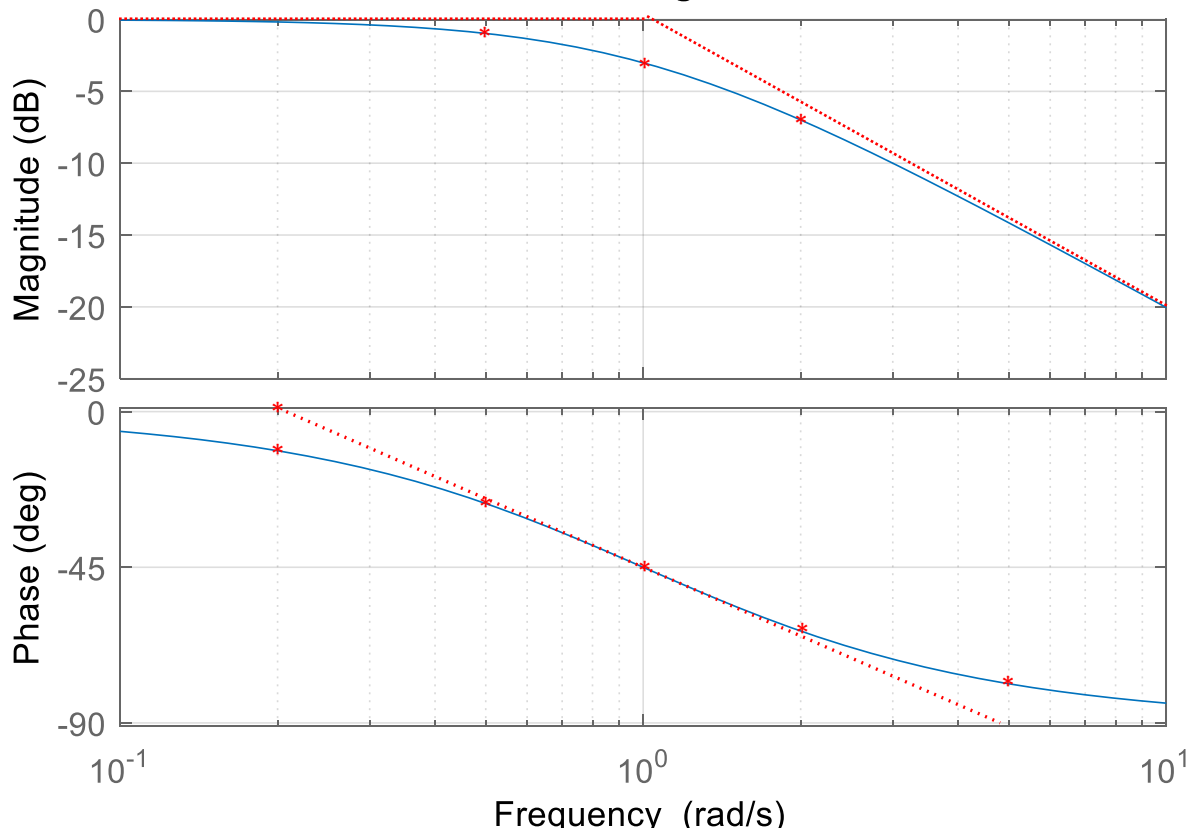
So, the slope, $m = \frac{d \arg T}{d \ln x} = -\frac{1}{2}$, at $x = 1$ (i.e. at $\omega = a$). This corresponds to a line that has $y = m \ln x + c$, going through $y = -\pi/4 = c$ at $\ln x = 0$, (i.e. $\omega = a$). This line goes through $y = 0$ at $m \ln x = c$,
 $x = e^{-\pi/2} = 0.2078$

Other points of support:

$x = \frac{\omega}{a}$	$\arg T(j\omega) = -\operatorname{atan} x$
0.1	-5.7° (~ -0.1 rad)
0.2	-11.3° (~ -0.2 rad)
0.5	-26.6°

$x = \frac{\omega}{a}$	$\arg T(j\omega) = -\operatorname{atan} x$
$10=1/0.1$	$-84.3^\circ = (-90^\circ + 5.7^\circ)$
$5=1/0.2$	$-78.7^\circ = (-90^\circ + 11.3^\circ)$
$2=1/0.5$	$-63.4^\circ = (-90^\circ + 26.6^\circ)$

Bode Diagram



For a quick sketch:

Magnitude: k_{dB} at low frequency, rolling off at -20 dB/dec at the corner frequency. Corrections: -1 dB below asymptote at $\omega/a = 0.5$, $\omega/a = 2$ (i.e. -7dB), and -3 dB at $\omega/a = 1$. Join the dots to get within ~ 1 dB.

Phase: Draw a line from $(\omega/a = 0.2, 0^\circ)$ to $(\omega/a = 5, -90^\circ)$. Mark points $\omega/a = 0.2, -11^\circ$ and $\omega/a = 5, -79^\circ$. Join the dots ... to get within $\sim 5^\circ$.

Second order low pass

$$T = \frac{1}{(s/\omega_n)^2 + 2\zeta s/\omega_n + 1}$$

$$\arg T(j\omega) = -\operatorname{atan}\left(\frac{2\zeta x}{1-x^2}\right), \text{ with } x = \omega/\omega_n$$

Slope of the argument with respect to $\ln x$ (as the Bode plot is drawn with respect to $\log \omega$)

$$\frac{d \arg T}{d \ln x} = \frac{d \arg T}{d x} \bigg/ \frac{d \ln x}{d x}$$

$$= -\left(\frac{2\zeta(1+x^2)}{(1-x^2)^2 + (2\zeta x)^2}\right) \bigg/ \left(\frac{1}{x}\right)$$

$$= -\frac{2\zeta(1+x^2)x}{(1-x^2)^2 + (2\zeta x)^2}$$

So, the slope, $m = \frac{d \arg T}{d \ln x} = -\frac{1}{\zeta}$, at $x = 1$ (i.e. at $\omega = \omega_n$). This corresponds to a line that has $y = m \ln x + c$, going through $y = -\pi/2 = c$ at $\ln x = 0$, (i.e. $\omega = \omega_n$). This line goes through $y = 0$ at $m \ln x = c$,

$$x = e^{-\zeta\pi/2} = (0.2078)^\zeta$$

Another point of support: $\arg T(j\omega) = -\operatorname{atan}\left(\frac{x2\zeta}{1-x^2}\right) = -\frac{\pi}{4} = -45^\circ$, at $x = \sqrt{1+\zeta^2} - \zeta$

The right hand two columns are due to symmetry of the phase plot. Coloured rows are shown on the figure below.

ζ	$x = e^{-\zeta\pi/2}$ arg = 0° intercept	$x = \sqrt{1+\zeta^2} - \zeta$ arg = -45°	$x = 1/(e^{-\zeta\pi/2})$ arg = -180° intercept	$x = 1/(\sqrt{1+\zeta^2} - \zeta)$ arg = -135°
0.1	0.85	0.91	1.17	1.10
0.2	0.73	0.82	1.37	1.22
0.4	0.53	0.68	1.87	1.48
0.5	0.46	0.62	2.19	1.62
0.701	0.33	0.52	3.04	1.93
1.0	0.21	0.41	4.81	2.41

```
d=[0.1;0.2;0.4;0.5;0.707;1],
round([exp(-d*pi/2),sqrt(1+d.^2)-d],2)
```

Small angle corrections:

$$\arg T(j\omega) = -\operatorname{atan}\left(\frac{x2\zeta}{1-x^2}\right) \approx -x2\zeta \text{ [rad] if } x \ll 1$$

$$\arg T(j\omega) = -\operatorname{atan}\left(\frac{x2\zeta}{1-x^2}\right) \approx -\pi + \frac{2\zeta}{x} \text{ [rad] if } x \gg 1$$

ζ	$x = 0.1$	$x = 0.2$
0.1	-1.2°	-2.4°
0.2	-2.3°	-4.8°
0.4	-4.6°	-9.5°
0.5	-5.8°	-11.8°
0.701	-8.1°	-16.4°
1.0	-11.4°	-22.6°

```
[atan(0.2*d*2/(1-0.2^2)),0.2*2*d] % Compare exact and approximation
```

For a quick sketch:

Magnitude: k_{dB} at low frequency, rolling off at -40 dB/dec at the corner frequency. Corrections:

Peak: $M(\omega_m) = k_{dB} - 20 \log_{10} 2\zeta \sqrt{1 - \zeta^2}$ at $\omega_m = \omega_n \sqrt{1 - 2\zeta^2}$

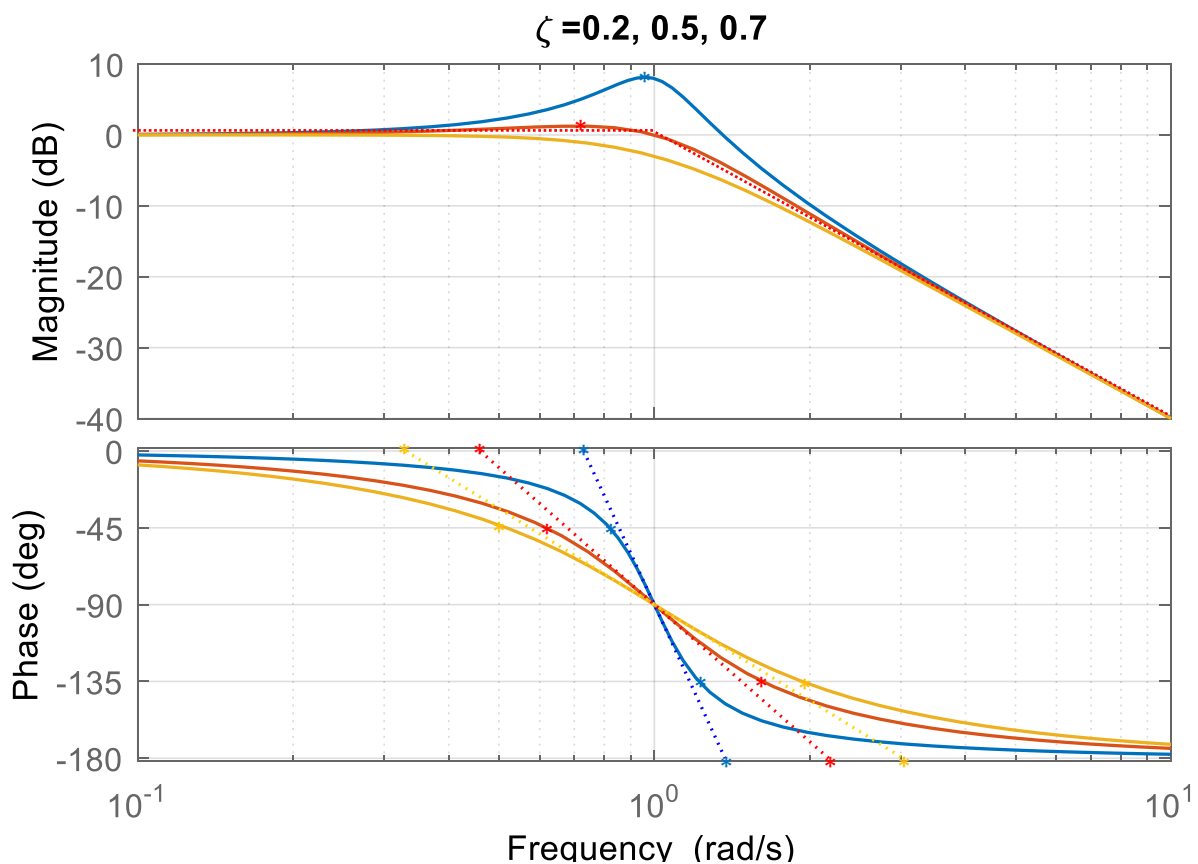
At corner frequency: $M(\omega_n) = k_{dB} - 20 \log_{10} 2\zeta$.

Goes through steady state gain at $\omega = \omega_n \sqrt{2(1 - \zeta^2)} = \omega_d \sqrt{2}$

Phase: Draw a line from 0° to -180° through $(\omega/\omega_n = 1, -90^\circ)$ and $(\omega/\omega_n = e^{-\zeta\pi/2}, 0^\circ)$

Corrections at -45° using $x = \sqrt{1 + \zeta^2} - \zeta$ (and -135°)

Small angle calculations: $x = 1/(1 + \sqrt{1 + \zeta^2} - \zeta)$. $\arg T(j\omega) = -\text{atan}\left(\frac{x2\zeta}{1-x^2}\right) \approx -x2\zeta$ [rad] if $x \ll 1$



```

Tp1=1/(s^2+2*0.1*s+1);
Tp2=1/(s^2+2*0.2*s+1);
Tp5=1/(s^2+2*0.5*s+1);
Tp7=1/(s^2+2*0.707*s+1);
bode(Tp2,Tp5,Tp7,{0.1,10}),title('\zeta =0.2, 0.5, 0.7')
hold on
plot([0.73,1/.73],[0,-180],'b:','LineWidth',1)
plot([0.46,1/.46],[0,-180],'r:','LineWidth',1)
plot([0.33,1/.33],[0,-180],'y:','LineWidth',1)

```